1 Security Model

(placeholder for paper.tex §4)

2 Evaluation

We evaluate the main circuit (v9) and its gate contract along the research questions of the study: relation conformance (Q1), budget conservation under concurrency (Q2), revocation (Q3), accountability (Q5) and cost (Q6), and compare against official implementations of neighbouring systems. **Every test reported here is a finite conformance test of the implementation against the stated relation; none of it is a security proof.** Security arguments are in Section 1; the non-cryptographic invariants that are machine-checked are listed in Section 2.8.

2.1 Setup

All measurements ran on one Apple M5 Pro (15 cores, 48 GiB), node 24.14, circom 2.2.3, snarkjs 0.7.6 (Groth16 over BN254), forge/anvil 1.5.1 (Prague). The v9 circuit has 60,622, 105,424, 150,226 and 195,028 constraints for $D = 1..4$ and 20 public inputs at $D = 4$ (27 before input compression). Phase 1 of the setup is the public Hermez ceremony (2^{18}) for the action, budget and revocation circuits, but a locally generated 2^{17} file for the registration circuit; *phase 2 is a single local contribution everywhere, so the keys are for evaluation only and a deployment needs a ceremony (for the registration circuit also a public phase 1).* Every experiment was pre-registered (attack lists, cases, statistics and targets committed before running); deviations are reported in the per-experiment records.

2.2 Relation conformance (Q1)

Mutation testing. Every single field of a valid witness was mutated and the circuit's verdict compared with the expected one: 199, 282, 408 and 534 field mutations plus 37–76 semantic cases for $D = 1..4$, with 0 unexpected outcomes. *Differential attack suite.* An independent plaintext oracle (separate Poseidon / keccak implementations) classifies 170 attack rows covering chain forgery, attenuation, revocation, registered roots, budget notes, the state root, the compressed accountability ciphertext and proof-level retargeting. Circuit and oracle agree on 114 rejections and 28 acceptances; 7 rows fail only because the attacker supplied a wrong path for a true statement; there are 0 completeness gaps. Of the 21 rows the circuit accepts and the oracle rejects, 20 are obligations of the contract by design (context hash, required capability, clock, root windows, replay) and each is closed by a contract test; the remaining one is the pre-registered witness-level row in which the holder legitimately proves a different context. An exhaustive single-field differential over all 534 fields at $D = 4$ finds 3 statement-level disagreements, exactly the three gate-bound inputs. At proof level, reusing a real proof with any changed action field, state root, budget nullifier or amount fails verification.

2.3 Budget conservation (Q2, H3)

We replayed concurrent spending with real transactions, all candidates of a race in one block. With per-edge caps only (v7 gate), 64 actions of 10 units under a budget of 100 were *all* accepted (640 units). With budget notes (v8/v9), 32 agents competing for the same budget obtained exactly 100 units, and of 12 actions built on the same note version only 4 (one per note) succeeded. Section 2.8 gives the machine-checked invariant behind these observations.

2.4 Revocation (Q3, H2)

In a shared world with chains that share prefixes (598 checks, 0 unexpected), revoking an edge or a key blocks exactly its descendants. The ablation that checks revocation only for the last hop accepts 14/14 descendants of a revoked edge, so the per-hop check is what blocks them. At scale (1,230 agents), revoking 1 % and then 10 % of the delegation edges (12 and 120 edges, 84 and 301 affected agents) left 0/0 descendants able to act and 0/0 unaffected agents blocked (witness-level check of every agent). A revocation costs 273,538 gas regardless of the size of the revocation set. The price of Δ-fresh revocation is availability: after the window, every in-flight proof built on the old root is rejected (64/64 and 64/64 in our samples), including those of unaffected agents, which re-prove once and succeed (32/32, 32/32).

2.5 Accountability (Q5, H4)

A 3-of-5 committee opens the accountability ciphertext with Chaum–Pedersen DLEQ partial decryptions and Lagrange interpolation in the group, without reconstructing the secret key (trusted dealer, static corruption of fewer than t members). All 65 pre-registered cases behaved as expected (0 deviations): exact opening for every 3-subset, identification of every malformed share, rejection of cross-event and wrong-epoch shares. A design comparison matters for the claim: with one ElGamal ciphertext per key (mode A), a wrong opening still produced valid curve points in 150/150 forced cases, so it cannot be detected; the KEM-plus-keystream design used by v9 (mode C) yields no valid output on a wrong key. The compressed v9 ciphertext is opened exactly in 10/10 fresh instances per depth, with a single key and with the threshold protocol.

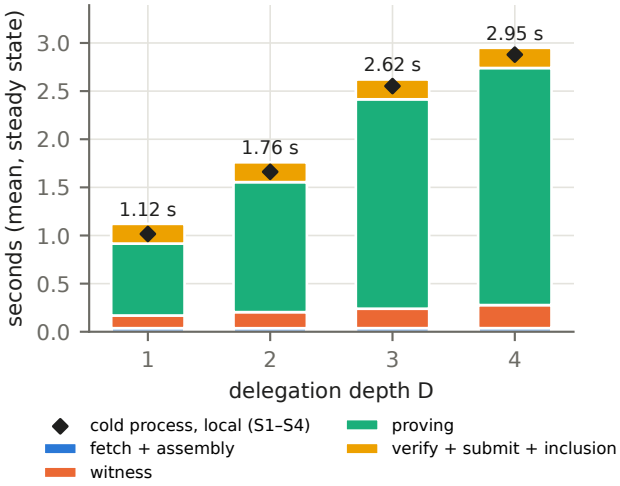


Figure 1: Lifecycle latency of one fresh action by stage (mean of 30 fresh actions per depth, steady state); diamonds: local computation in a fresh process. Proving dominates.

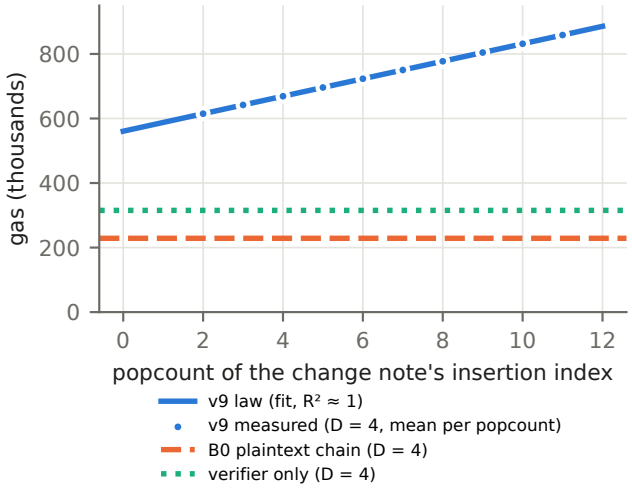


Figure 2: Execute gas at $D = 4$ against the number of one-bits of the change note's insertion index (1,024 actions), the fitted law, the plaintext lower bound B0 and the verifier alone.

2.6 Cost (Q6)

Latency. Fig. 1 breaks one fresh action into fetching paths, witness assembly and calculation, proving, off-chain verification, submission and inclusion. The local authorization computation takes 0.92, 1.55, 2.41 and 2.74 s for $D = 1..4$ (30 fresh actions per depth in three process blocks), of which proving takes 0.75–2.46 s; each hop adds 633 ms (95 % CI 627–641). A cold process adds 100–139 ms to the action. All three pre-registered targets hold at $D = 4$: p95 local computation 2.86 s (cold 2.94 s) ≤ 10 s, peak memory 4.51 GiB ≤ 8 GiB, verifier-only gas 315,215 $\leq$ 500k.

Gas. The full transaction cost of an action is set by the note-tree insertion, not by the depth (Fig. 2): gas $= 529{,}576 + 27{,}089 \cdot \text{popcount}(i) + 7{,}763 \cdot D$ ($R^2 = 0.999997$), where i is the insertion index of the change note. An action costs 564,147 gas on an almost empty note tree and 743,211 on average with 3,830 notes; a gas figure is therefore only meaningful together with the note-tree size.

Scale. 1,230 agents (1,024 at depth 4) with budgets delegated by real transfer transactions acted concurrently on the same state: 1,230/1,230 actions were accepted in 31 blocks of 30M gas (39.7 per block). One machine produced 23.1 proofs per minute.

Cost of privacy and of each component. A plaintext implementation of the same policy in the same contract stack (B0: ECDSA certificate chain, per-hop revocation, replay set, per-edge budget ledger) costs 115,215–228,705 gas for $D = 1..4$; v9 costs 2.45–4.66× that on a fresh note tree and 3.25–5.96× at the size of our world, with the ratio falling as D grows (the larger values at $D = 1$). In proving time, accountability adds 84 ms and budget notes add 63 ms at $D = 4$ (Fig. 3); at $D = 3$ accountability crosses a power-of-two FFT domain and adds 620 ms.

2.7 Comparison with neighbouring systems

We ran the official implementations natively instead of excluding systems built on other stacks, and compared only sub-tasks they support. UCAN (go-ucan) verifies a depth-4 delegation in 0.184 ms with 1,853 bytes of tokens, with the lineage in clear and no revocation in the implementation. The DAC scheme from equivalence-class signatures shows a credential in 2.73 ms and verifies it in 16.2 ms; its code supports a single delegation level. zk-creds proves an expiry show in 260 ms (one witness, repeated) and has no delegation. ZKM, ZK-ACE and ZEBRA have no public code and are compared at protocol level: ZKM shares hidden lineage, in-circuit conservation and on-chain verification but revokes only whole subtrees and offers only voluntary disclosure; ZK-ACE has no delegation. These systems do not offer the same functionality, so the timings are not a like-for-like comparison; what the comparison shows is the combination our system adds and its cost relative to the plaintext bound above.

2.8 Machine-checked invariants

A Lean 4 model of the gate (mint, split, delegate, execute, revoke, time) proves, for every reachable state, budget conservation (unspent plus spent equals minted), subtree conservation (below any certificate edge, spending never exceeds what was minted or delegated into it), monotonic revocation and freezing (after the window nothing below a revoked edge is spent): 6 theorem statements with 0

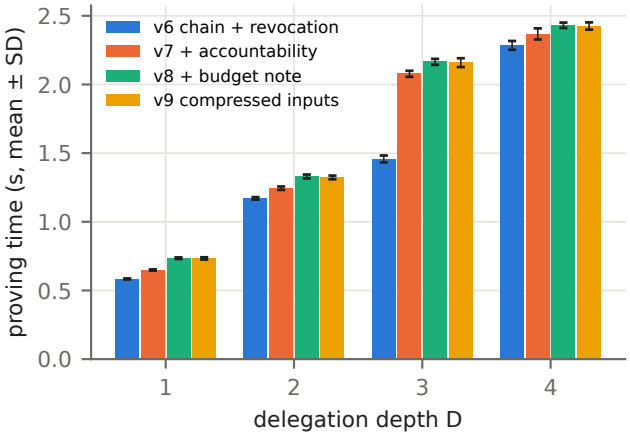


Figure 3: Proving time of successive circuit versions, 30 fresh instances per cell (mean ± SD).

uses of `sorry` and only Lean's standard axioms. The model abstracts the cryptography by stated assumptions (knowledge soundness, commitment binding, nullifier injectivity) and is checked for non-vacuity: removing the nullifier check lets the model spend twice its budget.

2.9 Limitations

One machine and one prover (snarkjs); native provers were not tested. The chain is a local anvil node, so inclusion latency excludes block time and finality. Phase 2 of the setup is not a ceremony. Tree heights are fixed by the compiled circuits and were not swept; depths beyond 4 were not compiled. The anonymity of v9 transcripts was not tested empirically, and the anonymity and no-false-framing arguments remain paper proofs.